Saeyoung Choi

Ph.D. Candidate in Seoul National University

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EDUCATION

Seoul National University, Seoul, South Korea — Mar 2023 – Present

- Technology Management, Economics and Policy Department, College of Engineering
- Ph.D. Candidate
- Thesis: Modeling Researchers' Trajectories with Deep Learning: Topic Search, Field Transition, and Career Adaptation (Advisor: Junghye Lee)

Yonsei University, Seoul, South Korea — Mar 2021 – Feb 2023

- M.S. in Mechanical Engineering
- Thesis: Thermodynamic analysis of solid oxide electrolysis cell system and its thermal integration (Advisor: Jongsup Hong)

Yonsei University, Seoul, South Korea — Mar 2015 – Feb 2021

- B.S. in Mechanical Engineering

RESEARCH INTERESTS

- Science & Technology Policy; Researchers' Career; Flexibility of Innovation System

RESEARCH METHODS

- Deep Learning on Sequential & Graph Data

PEER-REVIEWED PUBLICATIONS

1. **Choi, S.**, & Lee, J. (2026). Revealing country-level disparities in researchers' transition to artificial intelligence: A two-stage analytical framework, 20(3), 101843.
2. **Choi, S.**, & Hong, J. (2023). Versatile thermodynamic design of a solid oxide electrolysis cell system and its sensitivity to operating conditions. *Energy Conversion and Management*, *298*, 117776.
3. Min, G.*, **Choi, S.***, & Hong, J. (2022). A review of solid oxide steam-electrolysis cell systems: Thermodynamics and thermal integration. *Applied Energy*, *328*, 120145. (*: equally contributed)
4. Min, G., Park, Y. J., **Choi, S.**, & Hong, J. (2021). Sensitivity analysis of a solid oxide co-electrolysis cell system with respect to its key operating parameters and optimization with its performance map. *Energy Conversion and Management*, *249*, 114848.

WORKING PAPERS

- "Pivoting, not Leaping: Atypical Combinations with Anchored Exploration"
- "Who Survives the Scientific Pivot? Long-Term Consequences of Scientific Pivots"

CONFERENCES & WORKSHOPS

- **Saeyoung Choi**, Junghye Lee. "Pivoting, not Leaping: Atypical Knowledge Combinations with Anchored Exploration", International Conference on the Science of Science and Innovation (ICSSI) 2026, USA, June 2026 (Oral Presentation)
- **Saeyoung Choi**, Junghye Lee. "Who Survives the Scientific Pivot? Long-Term Consequences of Scientific Pivots", R&D Management Workshop 2026 in Seoul, Korea, May 2026 (Oral Presentation)
- 2026 Atlanta Academy on Science and Innovation Policy, USA, May 2026 [**Award**]
- **Saeyoung Choi**, Junghye Lee. "National Influence on Researchers' Topic Transitions: A Deep Learning Analysis", Korean Society for Innovation Management & Economics (KOSIME) 2025 Summer Conference, Korea, July 2025 (Oral Presentation) [**Outstanding Paper Award**]
- 2025 Summer Institute in Computational Social Science (SICSS)-Korea, Korea, June 2025

GRANTS & FELLOWSHIPS

- Humanities and Social Sciences Research Fellow (Type B), National Research Foundation of Korea (NRF), 2026–2027.

PATENTS

- J. Hong, G.B. Min, **S. Choi**, J. Lee, K. Jeong. “Multidimensional analysis system and method of fuel cell stack”. ***KR 10-2023-0087682*** (2023).
- J. Hong, **S. Choi**, J. Lee, K. Jeong. “Sequential-modular simulation apparatus and method for fuel cell system analysis”. ***KR 10-2023-0077377*** (2023).
- J. Hong, G.B. Min, **S. Choi**, J. Lee, K. Jeong, Y.J. Park. “Fluid stream property evaluation apparatus and method for a fuel cell system simulation”. ***KR 10-2023-0077376*** (2023).

INDUSTRY & OTHER EXPERIENCE

H-Cube Solutions, Seoul, South Korea — Nov 2021 – Dec 2022

- Lab spin-off (Technology Commercialization Project)
- Developed engineering software for fuel cell and electrolysis system design and analysis (source code for sequential simulation, GUI development)
- Prototype showcased at **CES 2023**
- Delivered software to industrial partners